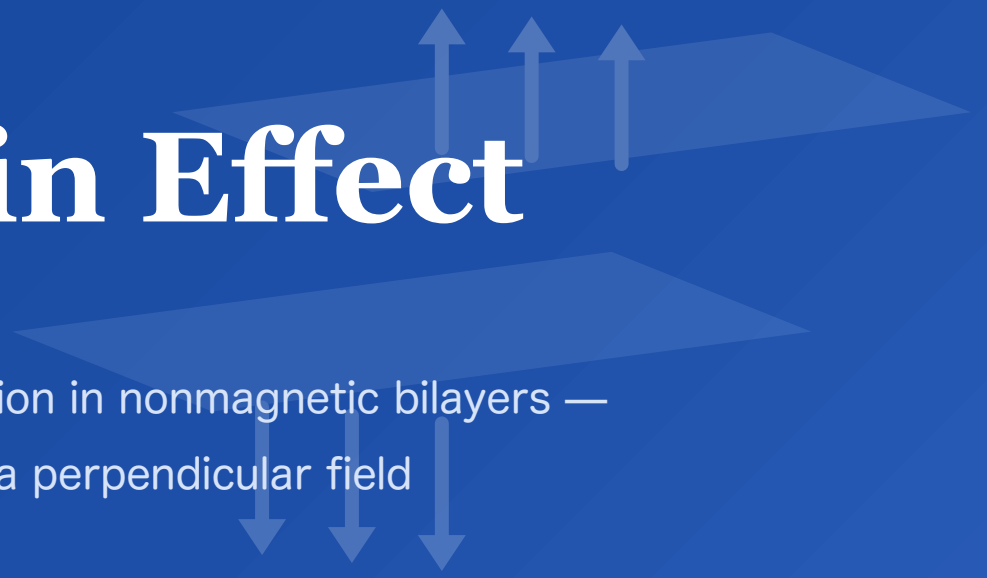
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Layer Edelstein Effect

Layer-resolved, current-induced spin magnetization in nonmagnetic bilayers — electrically generated and switched by a perpendicular field

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Literature Presentation

Background: The Route to the LEE

KNOWN EFFECT 1

Edelstein effect — a current induces spin polarization

- $\delta S = \chi \cdot j$ in noncentrosymmetric systems — verified in Rashba interfaces & topological insulators
- **Limitation: global response** — all layers summed, no layer addressability

+

KNOWN EFFECT 2

Layertronics — the layer index as a new degree of freedom

- Top- and bottom-layer electrons deflect oppositely — the layer Hall effect
- **Limitation: charge only** — no current-induced spin magnetization

↓

OPEN QUESTION

A current-induced spin magnetization that is **layer-resolved** and **switchable by an electric field**?

This Letter answers yes — by proposing the Layer Edelstein Effect (next page)

Core Proposal: The Layer Edelstein Effect (LEE)

Key concept

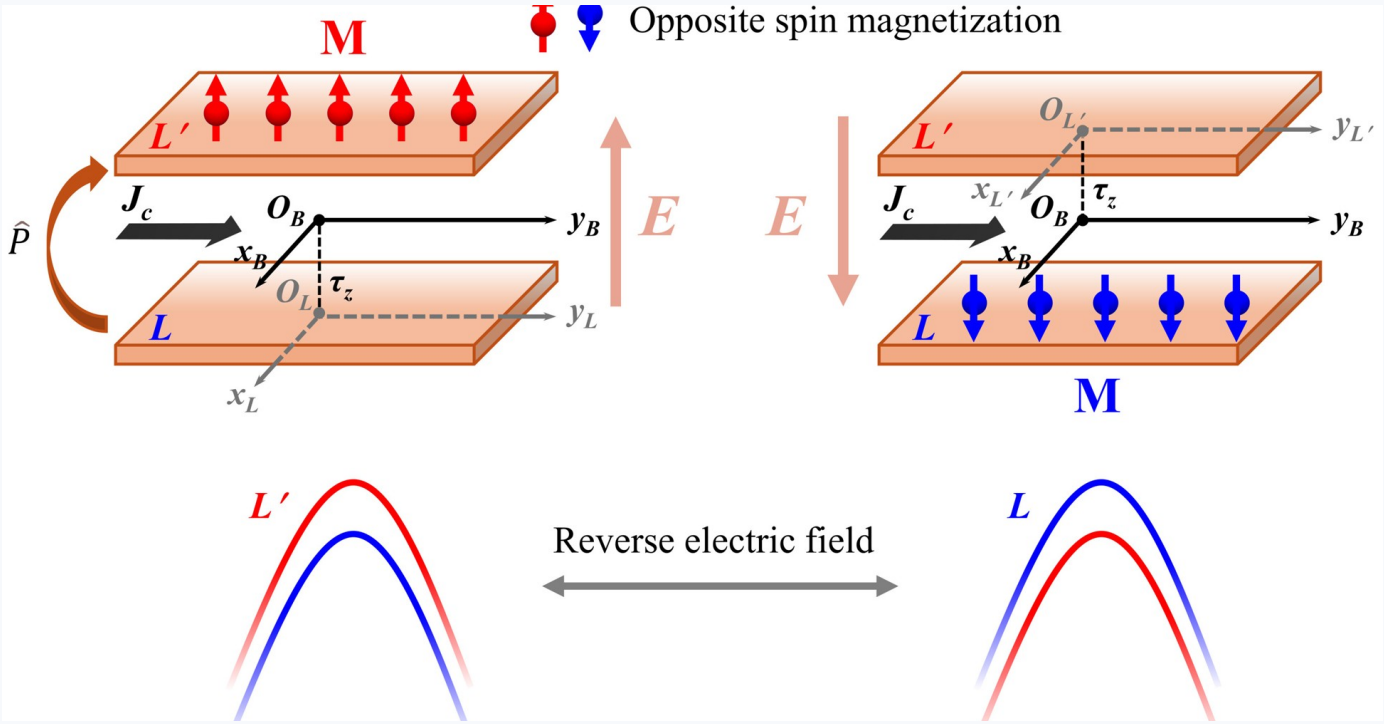


Fig. 1 — Stacking operator P̂ builds L′ from L; reversing the vertical field E flips the layer hosting the spin magnetization M

The LEE: a current drives opposite spin magnetizations on the two layers, and a perpendicular field selects which layer hosts them.

Layer-resolved counterpart of the Edelstein effect — the natural current-induced-spin member of the layer Hall effect family.

- In-plane current j_A induces layer-resolved spin magnetizations with opposite components.
- A perpendicular field E selects **which layer hosts the magnetism** and sets its direction.
- Allowed in both centrosymmetric and noncentrosymmetric bilayers.

E ↑ → spin-up M in the upper layer · E ↓ → spin-down M in the lower layer

Minimal k·p Model: Demonstrating the LEE

WHY THIS MODEL SUFFICES

· Symmetry dictates the tensor form — the model needs only the essential ingredients

· Two layers · Rashba SOC of **opposite sign** (τ_z factor) · field-tunable Δ

· Warping & spin mixing: renormalize magnitude only, never the tensor form (SM V)

$$H(\mathbf{k}) = \varepsilon_0\tau_0 \otimes \sigma_0 + t_\perp\tau_x \otimes \sigma_0 + \Delta\tau_z \otimes \sigma_0 + \alpha\,\tau_z \otimes (\sigma_x k_y - \sigma_y k_x)$$

τ — layer · σ — spin · Δ tunable by the out-of-plane field E

RESULT: THE RESPONSE TENSOR

$$\chi_{xy} = -\chi_{yx} \neq 0, \quad \chi \propto \alpha\Delta$$

· Δ = 0: inversion kept → exact cancellation, zero net

· Δ ≠ 0: weight → one layer; j_x (j_y) → M_y (M_x)

· Sign **reverses with Δ** — the LEE, demonstrated

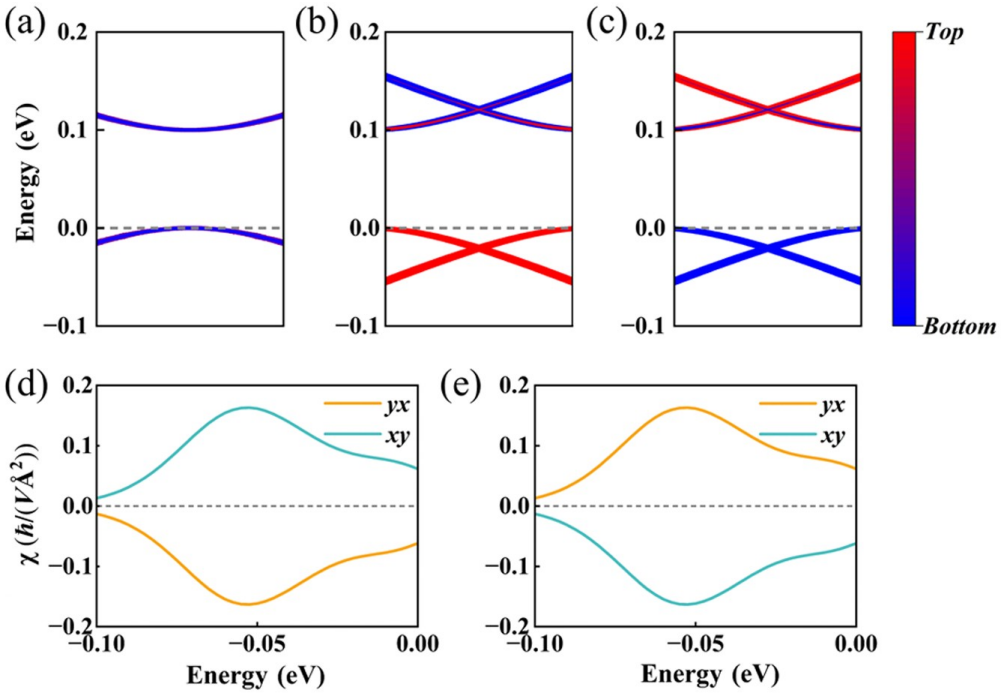


Fig. 2 — layer-selective bands under ±E and the induced response vs chemical potential

Symmetry Rules: How the Classification Is Computed

1 Enumerate all stackings

- Fix the monolayer layer group LG_L (80 types)
- Build every $P\hat{} = P\tau_z\tau_o$ from the quotient group LG_{lat} / LG_L, $LG_{lat} \in$ {C2h, D2h, D4h, D6h}

2 Build the bilayer point group

- Unified coordinates: ops extended to 4×4 $Q_{\pm}$
- Layer-keeping: $LG_B intra = LG_L \cap LG_{L'}$ · layer-exchanging: $LG_B inter = P'LG_L \cap LG_L P\hat{}^{-1}$
- Total LG_B = union of both subsets

3 Impose the tensor constraints

- Neumann's principle: $(R|\tau)$ constrains χ_{ij} exactly like $(R|0)$
- Fractional translations never matter — allowed χ_{ij} follows from the point group alone

4 Compare before vs after stacking

- Allowed in L but **newly forbidden in B** → Type I
- **All zero at zero field** → redo under E, track $LG_B inter$ → Type II

✓ Model-free verdict: the LEE type follows from the bilayer point group alone — no microscopic model required.

Two Manifestations of the LEE, Dictated by Symmetry

TYPE I Layer-opposite, field-redistributed

① Monolayer L noncentrosymmetric → χ_{ij}^{L} allowed

② Stacking symmetry (inversion / mirror) **forbids it in B**: $\chi_{ij}^{B} = 0$

③ Zero net → partner layer cancels L exactly:

$$\chi_{ij}^{L'} = -\chi_{ij}^{L}$$

E polarizes Fermi states onto **one layer** → sign & host layer flip with E

Typical: inversion/mirror stacking of noncentrosymmetric monolayers (Janus MoSSe → D3d / D3h)

VS

TYPE II Field-created response

① Stricter symmetry: χ_{ij}^{B}, χ_{ij}^{L}, $\chi_{ij}^{L'}$ all vanish at zero field (I, or M_z + C_{nz}, n > 2)

② Under E every LG_Binter op is broken — acting with one **equals reversing E** → components odd in E, layer-opposite

③ Condition: ≥ 1 component unconstrained by LG_Bintra

E's dual role: **creates** then **redistributes** the response

Typical: direct stacking of centrosymmetric monolayers ($MoTe_2$, $\hat{P} = \hat{E}$)

→ Both types follow one point-group rule — quantified on the next slide: **all 80 layer groups can host the LEE.**

A Model-Independent Symmetry Criterion

CONSEQUENCE OF TYPE I + TYPE II

80/80

Every layer group reaches the LEE via an appropriate stacking — the effect is **ubiquitous** in nonmagnetic bilayers.

Inversion stacking → noncentrosymmetric monolayers · direct translation stacking → centrosymmetric monolayers

Applying the same rule to all 80 layer groups: at least one LG_B inter op is essential — inversion does NOT forbid the LEE

Ci ✓ | C2h ✓ | D2h ✓ | C3i ✓ | D3d ✓ | C6h ✓

D4h ✓ | D6h ✓

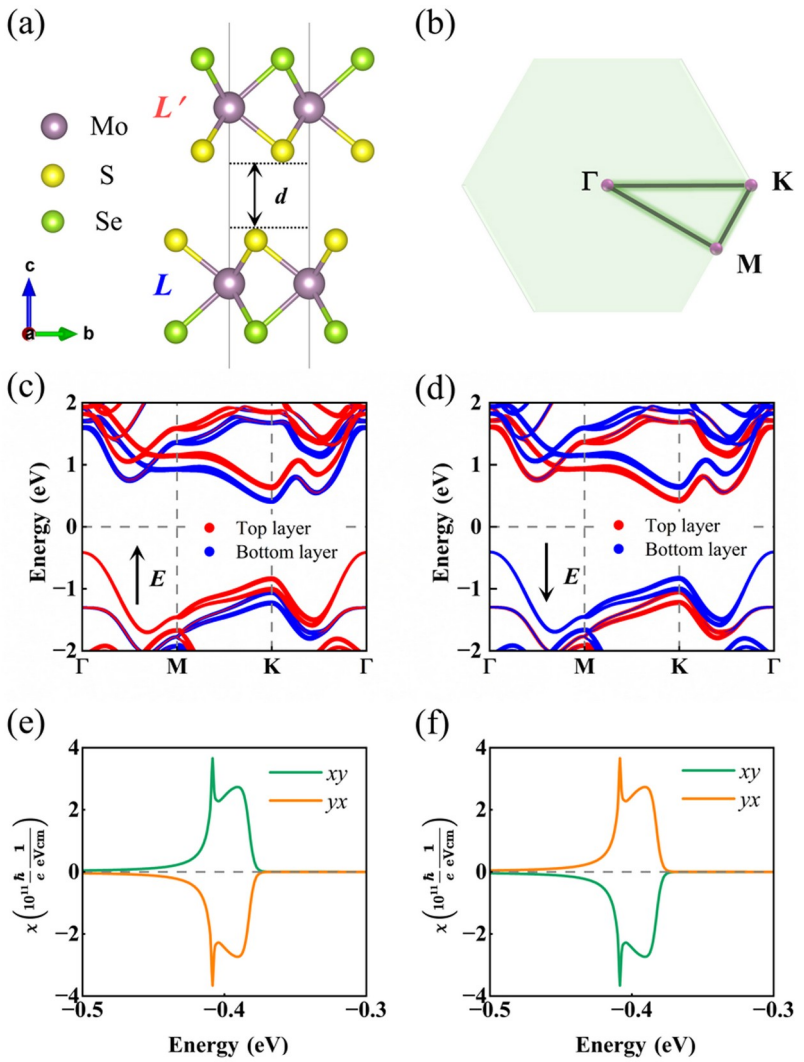
Centrosymmetric bilayer point groups — LEE allowed (representative entries of Table I)

C1 ✗ | C2(z) ✗ | C3 ✗ | C3v ✗ | C6v ✗

Some noncentrosymmetric groups also forbid it — the verdict is point-group only

Meaning: a **generic, symmetry-governed response** — candidate bilayers can be screened by group theory alone, before any DFT.

LEE in Realistic Materials: Semiconducting Janus MoSSe



SETUP

- Semiconducting Janus MoSSe monolayer, inversion stacking $P\hat{}= \hat{I} \rightarrow$ Type-I bilayer (D3d)

RESULTS (DFT)

- Under ±E, Fermi-level states localize onto **one single layer** (+E top / –E bottom) — Fig. 3c–d
- In-plane current j_x (j_y) induces M along **−y (x)** on the selected layer — Fig. 3e–f
- Reversing E flips the host layer and reverses $\chi_{xy} = -\chi_{yx}$

CONCLUSION

- Type-I LEE confirmed — finite-temperature calculations show it is **experimentally detectable** in dual-gated van der Waals devices

More Carriers: $MoTe_2$ and WTe_2 Bilayers

Same criterion, more platforms

The same symmetry criterion predicts a field-created LEE in centrosymmetric TMDs and extends unchanged to metallic systems.

SM Sec. IX · Sec. X

TYPE II Bilayer $MoTe_2$ — LEE switched on by the field

- Monolayer P6̄m2 carries $C_{3z} + M_z \rightarrow$ **all** Edelstein components vanish (Type II)
- Direct stacking ($\hat{P} = \hat{E}$) preserves P6̄m2 — yet the field induces $\chi_{xy} = -\chi_{yx} \neq 0$, opposite signs
- The field-created, field-controlled scenario in action

METALLIC 1T′ -WTe_2 — a direct metallic platform

- Stacked 1T′-WTe_2: a **metallic** realization of the LEE (SM Sec. X)
- No gate-defined carrier density required — closer to ambient transport measurements

Together with MoSSe, the predicted LEE spans **semiconducting and metallic** nonmagnetic bilayers — three concrete, synthesizable material candidates.

Device Perspectives

1

Electrical probing

A dual-gated van der Waals device supplies the vertical field and tunes the carrier density — the layer-resolved spin polarization can be read out through spin-to-charge conversion, with magnitude and sign tracked versus gate voltage (SM Sec. IX, XI).

MoSSe · $MoTe_2$ semiconducting platforms

2

Spin-orbit logic

The stacking degree of freedom opens a wide configuration space for designing spin-magnetization-switchable bilayer spin-orbit logic — distinct from the layer-dependent spin Hall effect, which shares the spin-layer coupling origin but serves different functionalities (SM Sec. XII).

Stacking as a design knob

3

Magnetic memory

The LEE uniquely switches between two opposite spin-magnetization configurations under an electric field — enabling all-electric manipulation of the "0" and "1" states in magnetic memory devices, without magnetic fields or magnetic ordering.

E-field writes the bit

→ From fundamental symmetry to working devices — the remaining step is experimental confirmation in stacked 2D materials.

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1. **The LEE — a new member of the Hall family**

   Layer-resolved, current-induced spin magnetization: the natural real-space counterpart of the layer Hall effect.

2. **Universal and symmetry-governed**

   A model-independent point-group criterion: at least one layer-exchange operation is essential — all 80 layer groups can realize it via proper stacking.

3. **Two complementary manifestations**

   Type-I explicit layer-opposite components and Type-II electric-field-induced components cover every nonmagnetic bilayer.

4. **Confirmed in realistic materials**

   First-principles verification on MoSSe, $MoTe_2$ and 1T′-WTe_2 bilayers — detectable in dual-gated van der Waals devices.

Thank You

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